Week 8

Case Study 1

Data for Questions 1 to 5

Insurance Claim Approval Prediction

An insurance analytics team builds a deep feedforward network to classify whether a health insurance claim needs manual review. Each record contains numerical features such as claim amount, hospital-risk score, patient-age group, policy duration, and number of previous claims. The output layer uses a sigmoid unit because the final output is a probability.

In the first version, the team also uses sigmoid activations in all eight hidden layers. Training starts normally but soon becomes very slow. The gradients reaching early layers become very small, especially when hidden pre-activation values are large in magnitude. The team replaces the hidden activations with ReLU:

$$\mathrm{ReLU}(x) = \max(0, x),$$

while keeping the sigmoid output layer. For analysis, they compare sigmoid and ReLU outputs at $x = -4$ and $x = 2$. Use $\sigma(2) \approx 0.88$ and $\sigma(-4) \approx 0.018$.

Question 1: Why can sigmoid hidden layers slow down training in a deep network?

A) They can saturate and produce very small gradients.

B) They always output negative values under the given conditions.

C) They remove all nonlinear behavior.

D) They force every weight to become zero.

Correct Answer: A

Explanation: Sigmoid units saturate for large positive or negative inputs, making derivatives small and causing vanishing gradients.

Question 2: What is $\mathrm{ReLU}(-4)$?

A) -4

B) -1

C) 0

D) 4

Correct Answer: C

Explanation: ReLU outputs zero for negative inputs, so $\max(0, -4) = 0$.

Question 3: What is ReLU(2)?

A) 0

B) -2

C) 1

D) 2

Correct Answer: D

Explanation: For positive inputs, ReLU returns the input itself.

Question 4: Which property makes ReLU helpful for positive pre-activation values?

A) It bounds hidden outputs between 0 and 1, as sigmoid would.

B) It avoids saturation and keeps a useful gradient.

C) It converts classification into clustering.

D) It makes the loss function unnecessary.

Correct Answer: B

Explanation: ReLU has derivative 1 for positive inputs, so gradients can flow more effectively.

Question 5: Why is the output layer still kept as sigmoid in this binary classifier?

A) It maps the final score to a probability-like value between 0 and 1.

B) It is required for every hidden layer for the described model.

C) It guarantees no validation error after training on every future test case.

D) It prevents all negative input features for the data shown here.

Correct Answer: A

Explanation: A sigmoid output is suitable for binary probability prediction.

Case Study 2

Data for Questions 6 to 10

Solar Power Forecasting for a Rooftop Plant

A renewable-energy company predicts next-hour solar generation from irradiance, temperature, cloud-cover index, panel age, and previous-hour generation. The network has seven hidden layers with ReLU activations. When weights are initialized with a variance of 1, activations in later layers become unstable and the loss oscillates. The team compares Xavier and He initialization:

$$\text{Xavier: } \operatorname{Var}(w) = \frac{1}{n}, \qquad \text{He: } \operatorname{Var}(w) = \frac{2}{n},$$

where n is the number of input connections to a neuron. Consider a hidden layer with $n = 200$ inputs.

Question 6: What problem can poor initialization create in a deep network?

A) It automatically removes all noise from data.

B) It can cause activations or gradients to explode or vanish.

C) It guarantees underfitting only under the given conditions.

D) It changes regression targets into class labels.

Correct Answer: B

Explanation: Poor initialization can make signal variance grow or shrink across layers, destabilizing training.

Question 7: What is the Xavier variance for $n = 200$?

A) 0.020

B) 0.200

C) 0.100

D) 0.005

Correct Answer: D

Explanation: Xavier variance is $1/200 = 0.005$.

Question 8: What is the He variance for $n = 200$?

A) 0.010

B) 0.005

C) 0.020

D) 0.100

Correct Answer: A

Explanation: He variance is $2/200 = 0.01$.

Question 9: Why is He initialization more suitable for these hidden layers?

A) The output layer uses no activation.

B) The network has only one layer.

C) The hidden layers use ReLU activations.

D) The target variable is binary in the stated case setting.

Correct Answer: C

Explanation: He initialization includes the factor of 2 used for ReLU-style activations.

Question 10: Which observation suggests the new initialization improved training stability?

A) Hidden activations are all forced to zero.

B) Loss decreases more smoothly across epochs.

C) No validation set is needed in the stated case setting.

D) All weights become identical.

Correct Answer: B

Explanation: Smoothly decreasing loss is evidence that the optimization is more stable.

Case Study 3

Data for Questions 11 to 15

Warehouse Sensor Fault Detection

A logistics company trains a deep network to detect abnormal warehouse conditions from temperature, vibration, humidity, door-event, and power-consumption sensors. The model has ten hidden layers. Direct end-to-end supervised training is slow, and the earliest layers learn weak representations because gradients reaching them are small.

The team adopts greedy layer-wise pretraining. First, the layer closest to the input is trained with an unsupervised reconstruction objective. Its weights are then frozen, and the next layer is trained using the previous layer's representation as input. This process continues until all hidden layers are pretrained. Finally, the output layer is added and the full network is fine-tuned using the labeled abnormal-condition task.

Question 11: What is the main goal of greedy layer-wise pretraining here?

A) To delete the hidden layers before training.

B) To avoid using input features during both unsupervised pretraining and fine-tuning.

C) To give each layer a useful initialization before supervised fine-tuning.

D) To guarantee zero test error in this specific scenario.

Correct Answer: C

Explanation: Layer-wise pretraining initializes hidden layers so supervised fine-tuning starts from useful representations.

Question 12: During pretraining of the second hidden layer, what happens to the already trained first layer?

A) It is randomly reinitialized in the stated case setting.

B) It is removed from the model.

C) It is trained with the output labels only.

D) It is kept fixed while the next layer is trained.

Correct Answer: D

Explanation: Greedy layer-wise pretraining freezes earlier trained layers before training the next layer.

Question 13: Why is the first pretraining objective described as unsupervised?

A) It uses the input data without using the abnormal-condition labels.

B) It uses no numerical features under the given conditions.

C) It uses only the final classifier output for the data shown here.

D) It requires the validation labels to be hidden in the stated case setting.

Correct Answer: A

Explanation: Reconstruction-based pretraining uses inputs as training signals rather than class labels.

Question 14: What happens after all hidden layers have been pretrained?

A) The network is discarded under the given conditions.

B) The full model is fine-tuned with the supervised task objective.

C) The input features are replaced by random values.

D) The loss function is no longer required in the stated case setting.

Correct Answer: B

Explanation: After pretraining, the complete network is trained end-to-end on the labeled objective.

Question 15: Which issue is layer-wise pretraining intended to reduce?

A) The need to store data files in this specific scenario.

B) The number of output classes under the case assumptions.

C) The difficulty of training early layers in a very deep network.

D) The existence of nonlinear activations under the given conditions.

Correct Answer: C

Explanation: Pretraining helps early layers learn useful representations before full supervised fine-tuning.

Case Study 4

Data for Questions 16 to 20

Laboratory Test Result Prediction

A diagnostic laboratory predicts whether a sample needs repeat testing based on numerical features such as machine confidence score, reagent age, sample temperature, processing time, and previous calibration drift. The network has six hidden layers. Without batch normalization, training is very sensitive to learning-rate changes because layer input distributions shift during training.

The team inserts batch normalization after each linear transformation and before ReLU. For one mini-batch, a neuron's pre-activation values are

$$1, \quad 3, \quad 5, \quad 7.$$

Ignore learnable scale and shift for the numerical questions.

Question 16: What problem does batch normalization primarily address?

A) It changes labels from binary to continuous.

B) It removes the need for backpropagation.

C) It guarantees a perfect classifier in the stated case setting.

D) It reduces shifting layer-input distributions during training.

Correct Answer: D

Explanation: Batch normalization stabilizes layer input distributions across mini-batches.

Question 17: What is the mean of the mini-batch values $1, 3, 5, 7$?

A) 3

B) 4

C) 5

D) 16

Correct Answer: B

Explanation: The mean is $(1 + 3 + 5 + 7)/4 = 4$.

Question 18: What is the batch variance using the average squared deviation?

A) 4

B) 6

C) 3

D) 5

Correct Answer: D

Explanation: The squared deviations are $9, 1, 1, 9$, so the variance is $20/4 = 5$.

Question 19: Where is batch normalization placed in this case?

A) Only after the output probability for the described model.

B) Before every input feature is collected.

C) Between the linear transformation and ReLU activation.

D) After the final prediction is thresholded.

Correct Answer: C

Explanation: The case places batch normalization after the linear operation and before ReLU.

Question 20: Why are learnable scale and shift parameters useful in batch normalization?

A) They allow the network to choose an appropriate normalized representation.

B) They delete the mini-batch statistics in this specific scenario.

C) They prevent gradients from being computed under the case assumptions.

D) They force every activation to remain exactly zero for the described model.

Correct Answer: A

Explanation: The scale and shift parameters let the network learn the representation that works best after normalization.